Synthesis: What the Results Taught Us

ECON 8310: Business Forecasting — Lecture 13

Department of Economics

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Lecture Outline

1 What We Built

2 What the Results Taught

3 Choosing a Method

4 Doing It Honestly

5 Where to Go Next

What We Built

Fourteen weeks, four families of method, and one dataset carried through all of them.

The Arc of the Course

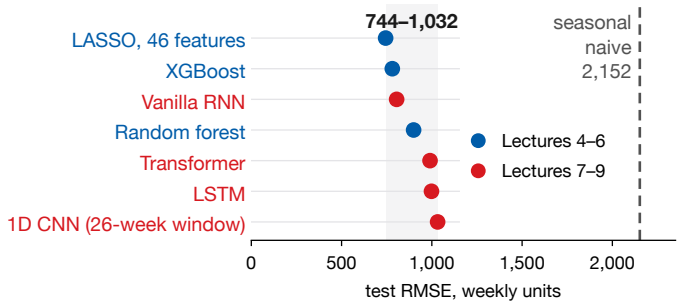
Part	Lectures	The central idea
Time series	1–3	Specify a structure — level, trend, season — and estimate it.
Trees	4–6	Stop specifying. Hand an algorithm features and let it find the structure.
Deep learning	7–9	Let the *architecture* encode what kind of structure to expect.
Bayesian	10–12	Change the deliverable: return a distribution, not a number.

Read down that last column and the course is not really four toolkits. It is four answers to a single question: **where does the knowledge in a forecast come from?**

From you, in the model structure. From the data, via a search procedure. From the architecture's assumptions. Or from a prior, stated openly and updated. **That framing is what survives when the libraries change.**

The Scoreboard

Every model in this course, on the M5 weekly panel — same 30 series, same held-out block.



Model	RMSE	Where
LASSO, 46 features	**744**	HW 4
XGBoost	781	HW 4
Vanilla RNN	805	L9 Part 1
Random forest	899	HW 3
Transformer	990	L9 Part 2
LSTM	998	L9 Part 1
1D CNN (26-wk)	1,032	L8
Seasonal naive	2,152	benchmark

XGBoost's 781 is from an earlier xgboost; the pinned 3.4.1 gives 792.

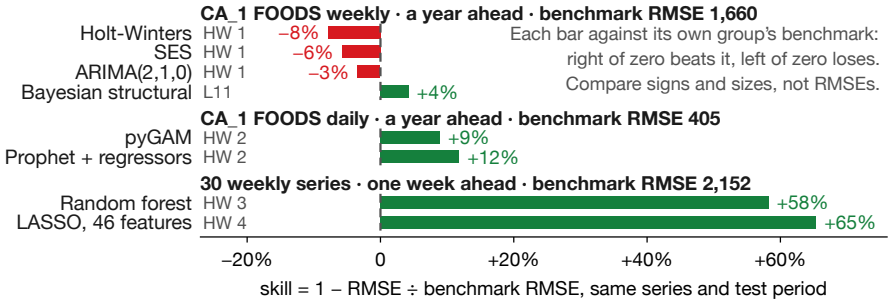
Not the ordering anyone predicts in Week 1. A 46-coefficient linear model is on top; the Transformer and the LSTM finished fifth and sixth of eight, behind a plain RNN with a fraction of their parameters, at the training budget each was measured at.

What the Results Taught

Three lessons that came out of measurements, not out of anyone's opinion.

Lesson 1: The Benchmark Is a Real Competitor

A year ahead, Homework 1's seasonal naive beat **every smoothing and ARIMA model the assignment asked for**. Holt-Winters estimated **52 seasonal indices from 4.3 annual cycles**; seasonal naive estimates **nothing**. Skill $= 1 - \text{RMSE} \div \text{benchmark RMSE}$.



Compute the benchmark first, every time. It costs one line and says whether the rest of the work was worth doing.

Lesson 2: More Sophisticated Did Not Mean Better

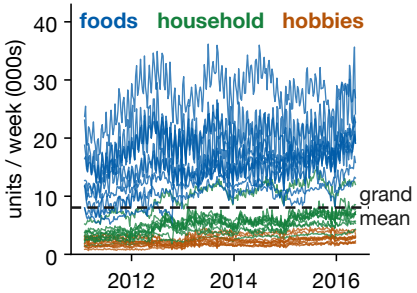
Four times this course, the more advanced method failed to win — and each time for a reason we could name.

LSTM lost to a plain RNN	At three windows, *at a fixed 30-epoch budget*; at 26 weeks they tie by 100 epochs. Gating fixes forgetting, and with 91% of the variance between series there is little to forget.
Transformer lost to an RNN	67,329 parameters against roughly **100 effectively independent** 52-week blocks — ours overlap 98%.
CNNs lost to LASSO	Their 26-week window held no annual lag; LASSO was handed one.
Tuning bought nothing measurable	Nine fits and a 3-unit change, against a **seed spread of 20–90** at one fixed config. CV improved; test did not.

None of these means the losing method is bad — each solves a problem this data does not have: long dependency chains, large corpora.

Lesson 2, Continued: A Problem This Data Does Not Have

Gates fix forgetting. What does this panel ask a model to remember? Lecture 9's 30 series:





between² / total² = 91%

sides: RMS deviation from the grand mean, units/week

91% of the variance is *which series*, not *which week*: little for a gate to keep. Transformers, likewise, earn their parameters on large corpora.

Match the method to the structure of the problem, not to the recency of the paper.

Lesson 3: One Idea Appeared Six Times

Lecture	Name it went by	What it penalized
3	GAM smoothing penalty	Wiggliness of a spline
4	Cost-complexity pruning	Number of leaves
5	`reg_lambda`, `gamma`	Leaf weights, leaf count
6	Ridge, LASSO, Elastic Net	Coefficient size
7	Weight decay, dropout	Network weights
11	Hierarchical prior	Distance from the group mean

Six model families that look nothing alike, one bargain underneath: **fit against complexity, with a single parameter setting the exchange rate.**

Five of those six tune that parameter by cross-validation. The sixth — the hierarchical prior — **estimates it**, which is the cleanest version of the idea in the course.

This is what to remember when the tools change. The libraries in this course will be replaced. The trade between fit and complexity will not.

Choosing a Method

A decision guide built from what we measured.

Which Method, When

If your situation is...	Start here
One series, clear seasonality, short history	ETS or ARIMA — and check the benchmark
Several seasonal cycles at once	GAM or Prophet
Many engineered features, tabular	**Regularized regression first**, then XGBoost
Many parallel series, shared drivers	Gradient boosting with series identifiers
Genuinely long dependencies, lots of data	Recurrent or attention architectures
Few observations per group	Hierarchical Bayesian — partial pooling
The decision needs a probability	Bayesian, for the predictive distribution

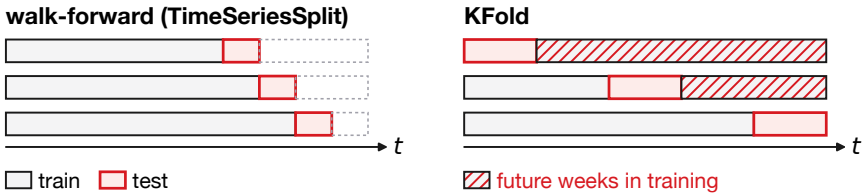
"Regularized regression first" is not a concession — it won our largest comparison and its coefficients can be read aloud. And the last row is the only one where the *question* changes rather than the accuracy.

Doing It Honestly

The errors in this field do not announce themselves.

Errors That Fail Silently I: the Pipeline

Confident, plausible, wrong and silent — and every one a **bug in the code**.



`KFold` on a time series	Future weeks train the model; CV looks great, the forecast fails.
Scaling outside the fold	Validation statistics leak into training.
Rolling features unshifted	The current week leaks the target into its own predictor.
Reading MDI importance	Noise ranked **17th of 29** in Homework 3 on distinct values alone.

Not a crash but a number that looks fine, so the discipline is procedural: split by time, scale inside the fold, shift the window, plant a control.

Errors That Fail Silently II: the Code Was Fine

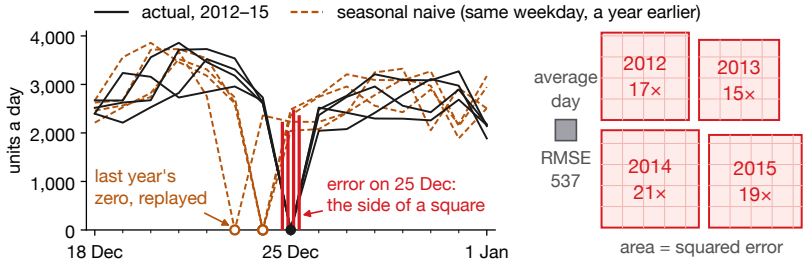
Harder, and more common in professional work. Here **nothing is wrong with the program**: the model fits, the diagnostics pass, and the answer is still wrong.

Pooling groups that are not **exchangeable**	Lecture 11 Part 2 pooled ten FOODS series with twenty that ignore SNAP. Textbook shrinkage, clean diagnostics, and it reported $+0.011$ where held-out weeks said $+0.089$.
Standardizing a coefficient you then **interpret**	Lecture 12: divide log price by its standard deviation and `b_price` stops being an elasticity. Identical fit, identical diagnostics, and the 5% price scenario moves by **22**$\times$.
Omitting a confounder	The naive elasticity was six times too large *with an interval half the width*. Precision is not accuracy.
Ignoring $\hat{R}$ and divergences	Our centered fits in Lecture 11 Parts 1 and 2 looked fine while failing $\hat{R}$ or divergences.

Only the last row has a diagnostic. The first three were caught by held-out data, by a DAG drawn before the model, and by asking what one unit of a variable means — none of which any library will run for you.

The Day Every Model in This Course Fails

Every 25 December the stores close and sales are **exactly zero**. CA_1 FOODS, 2012–15:



Four days, **0.25%** of the evaluation window, carry **4.6%** of the seasonal naive's squared error. Yet `is_event` fires on **158** days, only **5** of them closures.

> Knowable years ahead, and every method failed: the *encoding*, not the model, lumped a locked door in with the Super Bowl.

What a Defensible Forecast Report Contains

A benchmark	Computed, not cited. Every model measured against it.
A time-aware split	Walk-forward or a held-out final block, never random.
More than one evaluation	A single test split can flatter. Homework 4 ranked models the same way twice, which is why that ranking is believable.
The result that went against you	Including tuning that did not help, and the model you expected to win.
An interpretation	In the units of the business, not of the loss function.

The fourth row is the one that distinguishes professional work. A report where every result confirms the author's expectation is not a report anyone should act on.

You have practiced this all semester: the assignments were built so the obvious answer is frequently the wrong one, and the credit was for noticing.

Where You Do This Yourself

The final project asks for exactly the report on the previous slide, on data of your choosing.

Three methods, from at least two parts of the course	Time series, trees, deep learning, Bayesian. Three variations on one family do not count — the comparison must be between approaches differing *in kind*.
Plus a benchmark	It does not count toward the three, and it is not optional. Lesson 1 is why.
Walk-forward validation	For every comparison. Not a random split, not once.
A recommendation a non-specialist can act on	In the units of the business, with the uncertainty attached.
A methods reflection	Which model won, and what about *your* data made it win. This is where the course is actually assessed.

The last row is not a formality. Every scoreboard in this course came with a reason, and a project that reports a ranking without one has done the computation but not the work.

Where to Go Next

What we left out, and what to read.

What This Course Did Not Cover

Hierarchical reconciliation	Store forecasts that add up to regional totals (Hyndman and Athanasopoulos 2021, ch. 11).
Intermittent demand	Mostly-zero series: Croston's method and its relatives.
Structural breaks	The future stops resembling the past, and every method fails together.
Causal inference	DAGs told us confounders exist; credible causal estimates are a separate course.
Foundation models	Pretrained forecasters, zero-shot. FPP-Py Ch. 15 is a start.



Washington, DC, 18 March 2020.
Photo: Kurt Kaiser (CC0).

The third row is the one most likely to matter in your career. Every model here assumes the future resembles the past. When it stops, the model does not tell you — and the benchmark fails at the same moment.

Closing

1. **Compute the benchmark first.** It says whether the rest was worth doing.
2. **Match the method to the problem**: our most advanced finished fifth.
3. **One idea recurs:** fit against complexity, one parameter setting the rate.
4. **Split by time, always.** Leakage does not crash; it flatters.
5. **Diagnostics check the model, not the question.** Held-out data and a DAG caught our worst two.
6. **Report what went against you.**
7. **Know what the forecast is for.** Decisions need a distribution.

The libraries will change. The habits are the transferable part.

References I



Hyndman, Rob J. and George Athanasopoulos (2021). *Forecasting: Principles and Practice*. 3rd. Open access. Melbourne, Australia: OTexts. URL: https://otexts.com/fpp3/.